NMC 2015 A-LEVEL PAPER 1 QUESTIONS

SECTION A

Qn1. Let a, b and c be real numbers such that $(a-6)^2+(b-4)^2+(c-5)^2=0$. Find the value of $a+b+c$.

Qn2. Let S_n and T_n be the respective sums of the first n terms of two arithmetic series. If $S_n : T_n = (7n+1) : (4n+27)$ for all n. Find the ratio of the eleventh term of the first series to the eleventh term of the second series.

Qn3. Find the number of terms in the simplified expansion of $\left((a+3b)^2(a-3b)^2\right)^9$.

Qn4. Find the sum of the maximum and the minimum values of $\sin^6 x + \cos^6 x$.

Qn5. Prove that if a, b and c are distinct real numbers such that $\frac{a}{b-c}+\frac{b}{c-a}+\frac{c}{a-b}=0$, then

$$\frac{a}{(b-c)^2}+\frac{b}{(c-a)^2}+\frac{c}{(a-b)^2}=0.$$

Qn6. Find the number of pairs (x, y) of non-zero real numbers such that three of $x+y, x-y, xy$ and $\frac{x}{y}$ have the same value.

Qn7. Daniel has 3 red socks, 3 yellow socks and 3 blue socks. If he picks them without looking, what is the smallest number he must take in order to guarantee that he has four socks that form two pairs of socks of matching colours?

Qn8. Which numbers between 20 and 50 are exactly divisible by $3^{24}-1$?

Qn9. A mathematical challenge book contains 213 problems from 12 years of competitions. At first there were 25 problems per year followed by 16 problems per year, and finally 20 per year. Find the number of years when there were 25 problems.

Qn10. Initially, 2014 is written on a blackboard. At any time, we may write a new integer on the blackboard, which is either 1 greater than the existing integer, or half of the existing even integer. Find the minimum number of integers on the blackboard when the integer 1 appears.

SECTION B

Qn11. Find all polynomials $P(x)$ that satisfy the equation $P(x^2)+2x^2+10x=2xP(x+1)+3$.

Qn12. Suppose we perform the following four operations on a number h. Add 2, subtract 2, multiply by 2 and divide by 2, one after another, each exactly once, but not necessarily in that order of operation. With a clear explanation of the choice of sequence of these four operations on h, what is the largest number that we can end up with?

Qn13. An operation denoted by $*$ defines, for each pair of numbers (x, y), a number $x*y$ so that for all x, y and z, the identities $x*x=0$ and $x*(y*z)=(x*y)+z$ hold. (Note that $+$ denotes ordinary addition of numbers). Find $2015*2000$.

Qn14. Suppose x, y and z are integers satisfying the following conditions.

(i) $x - y + z = 3$ (ii) $x > 10$ and $y \geq 8$ (iii) $y^2 + z^2 < 80$.

What is the largest value of x?

Qn15. (a) K, L, M and N are points on sides AB, BC, CD and DA, respectively, of the unit square $ABCD$ such that KM is parallel to BC and LN is parallel to AB. The perimeter of triangle KLB is equal to 1. What is the area of triangle MND?

(b) In triangle $ABC, AB = AC$. A line L is drawn through A parallel to BC. A circle of radius $1cm$ is drawn tangent to the line L, the extended line BC and to the circle inscribing triangle ABC. Determine the radius of the circle inscribing the triangle ABC.

NMC 2015 A-LEVEL PAPER 1 SOLUTIONS

SECTION A

Qn1. Since each of the squares is greater or equal to zero, the sum of these squares will be zero only if each of the squares is zero. Hence $(a-6)^2 = 0, (b-4)^2 = 0$ and $(c-5)^2 = 0$. This implies that $a = 6$, $b = 4$ and $c = 5$. Thus $a + b + c = 6 + 4 + 5 = 15$.

Qn2. Let a_1, a_2 denote the first terms and d_1, d_2 common difference in the Arithmetic Progressions with n^{th} sums S_n and T_n respectively. Then for all n,

$$\frac{S_n}{T_n} = \frac{\frac{n}{2}[2a_1 + (n-1)d_1]}{\frac{n}{2}[2a_2 + (n-1)d_2]} = \frac{2a_1 + (n-1)d_1}{2a_2 + (n-1)d_2} = \frac{7n+1}{4n+27}.$$

The 11th terms of these series are $u_{11} = a_1 + 10d_1$ and $v_{11} = a_2 + 10d_2$ respectively. The ratio

$$\frac{u_{11}}{v_{11}} = \frac{a_1 + 10d_1}{a_2 + 10d_2} = \frac{2a_1 + 20d_1}{2a_2 + 20d_2}$$

is an expression for $\frac{S_n}{T_n}$ for $n = 21$. Thus $\frac{u_{11}}{v_{11}} = \frac{7(21)+1}{4(21)+27} = \frac{148}{111} = \frac{4}{3}$.

Qn3. The binomial expansion of $(a+b)^n$ has $n+1$ terms. Hence we have $((a+3b)^2(a-3b)^2))^9 = ((a^2-9b^2)^2))^9 = (a^2-9b^2)^{18}$. Therefore the expansion has $18 + 1 = 19$ terms.

Qn4.

$$\begin{aligned}
(\sin^2 x + \cos^2 x)^3 &= \sin^6 x + 3\sin^4 x \cos^2 x + 3\sin^2 x \cos^4 x + \cos^6 x \\
(1)^3 &= \sin^6 x + 3\sin^4 x \cos^2 x + 3\sin^2 x \cos^4 x + \cos^6 x \\
\sin^6 x + \cos^6 x &= 1 - 3\sin^4 x \cos^2 x - 3\sin^2 x \cos^4 x \\
&= 1 - 3\sin^2 x \cos^2 x (\sin^2 x + \cos^2 x) \\
&= 1 - 3\sin^2 x \cos^2 x \\
&= 1 - \frac{3}{4}(2\sin x \cos x)^2 = 1 - \frac{3}{4}\sin^2 2x.
\end{aligned}$$

The maximum value is 1 and occurs when $x = 0^\circ$ and the minimum value is $\frac{1}{4}$ and occurs when $x = 45^\circ$. The desired sum is $1 + \frac{1}{4} = \frac{5}{4}$.

Qn5. We have that

$$\begin{aligned}
0 &= \left(\frac{a}{b-c} + \frac{b}{c-a} + \frac{c}{a-b}\right)\left(\frac{1}{b-c} + \frac{1}{c-a} + \frac{1}{a-b}\right) \\
&= \frac{a}{(b-c)^2} + \frac{b}{(c-a)^2} + \frac{c}{(a-b)^2} + \frac{b+c}{(c-a)(a-b)} + \frac{c+a}{(a-b)(b-c)} + \frac{a+b}{(b-c)(c-a)} \\
&= \frac{a}{(b-c)^2} + \frac{b}{(c-a)^2} + \frac{c}{(a-b)^2} + \frac{b^2-c^2+c^2-a^2+a^2-b^2}{(a-b)(b-c)(c-a)} \\
&= \frac{a}{(b-c)^2} + \frac{b}{(c-a)^2} + \frac{c}{(a-b)^2} + \frac{0}{(a-b)(b-c)(c-a)}.
\end{aligned}$$

Hence $\frac{a}{(b-c)^2} + \frac{b}{(c-a)^2} + \frac{c}{(a-b)^2} = 0$ as required.

Qn6. Clearly $x + y \neq x - y$. We thus have $xy = \frac{x}{y} \implies xy^2 - x = 0$. So, $x(y^2 - 1) = 0$. Since x is non-zero, we must have that $y^2 - 1 = 0$ so that $y = \pm 1$. The only valid solution is $y = -1$. If $y = -1$, since $x - y = xy$, we have $1 + x = -x \implies x = \frac{-1}{2}$. Also from $x + y = xy$ we have $x - 1 = -x$ so that $x = \frac{1}{2}$. Hence the possible pairs are $(x, y) = (\frac{1}{2}, -1)$ and $(\frac{-1}{2}, -1)$.

Qn7. If Daniel picks 5 socks, he may have 3 red socks, 1 yellow and 1 blue or 1 red , 3 yellow and 1 blue socks or 1 red, 1 yellow and 3 blue socks. Each of these forms only one matching pair. If Daniel picks out 6 socks and leaves out 3 socks, he can leave out at least 2 socks of at most one colour. Hence of the six socks he takes, he can guarantee to have 2 matching pairs.

Qn8. $3^{24} - 1 = (3^{12} - 1)(3^{12} + 1) = (3^6 - 1)(3^6 + 1)(3^{12} + 1) = (3^3 - 1)(3^3 + 1)(3^6 + 1)(3^{12} + 1) = (26)(28)(3^6 + 1)(3^{12} + 1)$. Therefore $3^{24} - 1$ is a multiple of 26 and 28. Thus 26 and 28 are divisors of $3^{24} - 1$ that are between 20 and 50.

Qn9. Let x, y and z denote the respective numbers of years in which there were $25, 16$ and 20 problems. Then we have that $x + y + z = 12$ and $25x + 16y + 20z = 213$. Solving the two equations simultaneously and eliminating y gives $9x + 4z = 21$. Since both x and z are integer values, the equation $9x + 4z = 21$ has a solution only when $x = 1$ and $z = 3$. Therefore it is only one year that contained 25 problems.

Qn10. Since the objective is to downsize 2014 to 1, we should be halving as much as we can and increasing only when necessary. It follows that the most efficient sequence is $2014, 1007, 1008, 504, 252, 126, 63, 64, 32, 16, 8, 4, 2, 1$. Thus there are 14 numbers.

SECTION B

Qn11. $P(x)$ cannot be a constant polynomial since this would imply that the left hand side of the equation is a quadratic polynomial while the right hand side is linear.

Let $P(x)$ be of degree $n \geq 1$. Then the left hand side is of degree $2n$ and the right hand side degree $n+1$. We must have $2n = n+1$ which implies that $n = 1$. Hence $P(x)$ is of the form $ax + b$. Therefore

$$P(x^2) + 2x^2 + 10x = ax^2 + b + 2x^2 + 10x = 2x(a(x+1) + b) + 3 = 2ax^2 + 2(a+b)x + 3.$$

Comparing the constant terms, we have that $b = 3$.

Comparing the linear terms, we have $10 = 2a + 2b \Longrightarrow a = \frac{10-6}{2} = \frac{4}{2} = 2$.

Finally we check that the quadratic terms also agree. That is $a + 2 = 2a$ holds, which is the case. Hence the only polynomial that satisfies the functional equation is $P(x) = 2x + 3$.

Qn12. The following rules should be applied. $+$ and $-$ should not follow each other. Similarly, $\div$ and $\times$ should not follow each other as both would lead to the same number h we started with.

$+$ should preceed (come before) $\times$ since $(h+2)2 > 2h + 2$. Also $\div$ should preceed $+$ since $\frac{h}{2} + 2 > \frac{h+2}{2}$. This means that the order of the three signs is $\div, +, \times$.

It now remains to place $-$ either before $\div$ or after $\times$. Since $2h - 2 > (h-2)2 = 2h - 4$, it means $\times$ should preceed $-$. Since $\frac{h-2}{2} > \frac{h}{2} - 2$, then $-$ should preceed $\div$. Therefore the order of the signs is $-, \div, +, \times$ or $\div, +, \times, -$.

By comparing the numbers formed, $(\frac{h}{2} + 2)2 - 2 = h + 4 - 2 = h + 2$ and $\left(\frac{h-2}{2} + 2\right) 2 = h + 2$. The two numbers are the same, so $h + 2$ is the largest number we end up with.

Qn13. We have the following identities.

$$x * x = 0 \tag{1}$$
$$x * (y * z) = (x * y) + z \tag{2}$$

Put $x = y = z$ in equation (2). Then $x * (x * x) = (x * x) + x$.

From Equation 1, we have $x * 0 = 0 + x = x$ for all x.

Put $y = z$ in equation (2). Then $x * (y * y) = (x * y) + y$.

By equation (1), we have $x * 0 = (x * y) + y \Longrightarrow x * y = x - y$ for all x and y.

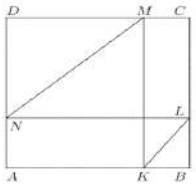
Therefore $2015 * 2000 = 2015 - 2000 = 15$.

Qn14. (i) $x - y + z = 3$ $\quad$ (ii) $x > 10$ and $y \geq 8$ $\quad$ (iii) $y^2 + z^2 < 80$.

Firstly from $y \geq 8$, we obtain that $y^2 \geq 64$. Using the inequality $y^2 + z^2 < 80$, we have that $z^2 < 80 - y^2 < 80 - 64 = 16$. That is $z^2 < 16$ and so $z \in \{-3, -2, -1, 0, 1, 2, 3\}$.

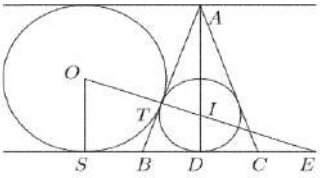
In addition, since $64 \leq y^2 < 80$ and y is an integer, then we must have that $y = 8$. But from equation (i), $10 < x = 3 + y - z = 3 + 8 - z \leq 11 - (-3)$ just by taking the lowest possible value of z, i.e $10 < x \leq 14$. Hence $x \leq 14$. The largest possible value of x is 14.

Qn15. (a) Let $AK = x$ and $AN = y$.



From $1-(1-x)-y = 1-KB-LB = KL = \sqrt{(1-x)^2+y^2}$, we have $x^2-2xy+y^2 = 1-2x+x^2+y^2$, which simplifies to $2x-2xy = 1 \Longrightarrow 2x(1-y) = 1$. Hence the area of triangle DMN is equal to $\frac{1}{2}DM \times DN = \frac{1}{2}x(1-y) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

(b) Let O be the centre of the larger circle and let this circle be tangent to BC at S. Let T be the point of common tangency of the two circles, and let I be the centre of the circle inscribing triangle ABC. Since $AB = AC$, I lies on the altitude AD and D is also the point of tangency of the incircle with BC.



Extend OI to meet BC at E. Let the inradius of ABC be r. Since $OS = 1$, then $AD = 2$. Triangles BAD and BET are congruent since $< ABD =< EBT$.

Now $< ADB = 90^\circ =< ETB$ and $BD = BT$. Hence $ET = 2$ and $EI = 2-r$. Now triangles SOE and DIE are similar.

Hence $r = \frac{ID}{OS} = \frac{IE}{OE} = \frac{2-r}{3}$. It follows that $3r = 2-r \Longrightarrow r = \frac{1}{2}$.

NMC 2015 A-LEVEL PAPER 2 QUESTIONS

SECTION A

Qn1. Find the positive value of M for which $22222^2 \times 55555^2 = 10^2 \times M^2$?

Qn2. While Annet was reading a novel of less than 1000 pages, she noticed that the sum of the digits on the page she was reading was 19 and the sum of the digits on the next page was 2. What is the number of the page that Annet was reading?

Qn3. The sum of the two 5-digit numbers UMS14 and UMS15 is 123429. What is the value of U+M+S?

Qn4. Nineteen students, numbered 1 through 19 stood in a circle in a clockwise numerical order, all facing the centre. They began to count out loud in a clockwise order: The first called out the number 1, the second called out 2 and each student then called out the number 1 more than the number called to his right. What was the number of the student who called out 2015?

Qn5. Whenever brothers A, B and C are asked a question, TWO lie and ONE tells the truth. I asked who amongst them was the youngest. A said that he was not the oldest. B replied that C was the oldest while C said that A was the youngest. Who actually was the youngest among the three brothers?